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ELECTROCHEMICAL STACK

Abstract

An electrochemical stack includes a water passage member interposed between two electrochemical cells among a plurality of electrochemical cells, the two electrochemical cells being located in a central region in a stacking direction of the electrochemical cells. The water passage member is provided with a water introduction unit and a flow path. The water introduction unit introduces water supplied from the outside. The flow path allows water introduced from the water introduction unit to flow along the electrochemical cells and guides the water to a water introducing communication passage penetrating the plurality of electrochemical cells in the stacking direction of the electrochemical cells.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-029405 filed on Feb. 29, 2024, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

FIELD OF THE INVENTION

[0002] The present disclosure relates to an electrochemical stack.

DESCRIPTION OF THE RELATED ART

[0003] There are electrochemical stacks including stacked electrochemical cells. Examples of the electrochemical stack include a water electrolysis stack, a hydrogen compression stack (an electrochemical hydrogen pump stack), and a fuel cell stack. In the water electrolysis stack, a plurality of water electrolysis cells are stacked. In the hydrogen compression stack, hydrogen compression cells are stacked. In the fuel cell stack, a plurality of unit cells are stacked.

[0004] JP 2019-123899 A discloses a water electrolysis system including a water electrolysis stack. In this water electrolysis system, water is circulated so that water stored in the gas-liquid separator is supplied into the water electrolysis stack, and unreacted water that has not been electrolyzed in the water electrolysis stack is discharged to the gas-liquid separator. The water supplied into the water electrolysis stack is supplied to each of the plurality of water electrolysis cells. Unreacted water that has not been electrolyzed in each of the plurality of water electrolysis cells is supplied from the water electrolysis stack to the gas-liquid separator.

SUMMARY OF THE INVENTION

[0005] The electrochemical stack generates heat during operation, and thus the temperature of the electrochemical stack increases. The temperature distribution within the electrochemical stack is not uniform and varies. The progress of deterioration of the electrolyte membranes provided in the electrochemical cells having a relatively high temperature tends to be faster than the progress of deterioration of the electrolyte membranes provided in the electrochemical cells having a relatively low temperature. Therefore, it has been desired to cool the electrochemical cells having a relatively high temperature.

[0006] The present invention has the object of solving the aforementioned problem.

[0007] An aspect of the present disclosure is an electrochemical stack comprising: a plurality of electrochemical cells stacked on one another; a water introduction unit configured to introduce water supplied from an outside; a water lead-out unit configured to lead out the water to the outside; a water introducing communication passage penetrating the plurality of electrochemical cells along a stacking direction of the electrochemical cells, and configured to guide the water to a first port provided in each of the plurality of electrochemical cells; a water lead-out communication passage penetrating the plurality of electrochemical cells along the stacking direction, and configured to guide, to the water lead-out unit, the water discharged from a second port provided in each of the plurality of electrochemical cells; and a water passage member interposed between two of the plurality of electrochemical cells, the two electrochemical cells being located in a central region in the stacking direction, wherein the water passage member is provided with the water introduction unit, and a flow path through which an entire amount of the water introduced from the water introduction unit flows along the electrochemical cells and is guided to the water introducing communication passage.

[0008] According to the aspect of the present disclosure, it is possible to increase the degree of cooling of the electrochemical cells in the end regions, and the electrochemical cells in the central region which have a relatively high temperature.

[0009] The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a diagram showing a water supply system according to a first embodiment;
[0011] FIG. 2 is a schematic diagram showing an electrochemical stack according to the first embodiment;
[0012] FIG. 3 is a diagram showing the water supply system according to a second embodiment;
[0013] FIG. 4 is a schematic diagram showing the electrochemical stack according to the second embodiment;
[0014] FIG. 5 is a schematic diagram showing the configuration of a water passage member;
[0015] FIG. 6 is a diagram showing the flow of water in a water circulation flow path during a first circulation operation;
[0016] FIG. 7 is a diagram showing the flow of water in the electrochemical stack during the first circulation operation;
[0017] FIG. 8 is a diagram showing the flow of water in the water circulation flow path during a second circulation operation;
[0018] FIG. 9 is a diagram showing the flow of water in the electrochemical stack during the second circulation operation;
[0019] FIG. 10A is a schematic diagram showing the temperature distribution of an electrochemical cell in a case where only the first circulation operation is performed;
[0020] FIG. 10B is a schematic diagram showing the temperature distribution of the electrochemical cell in a case where only the second circulation operation is performed; and
[0021] FIG. 10C is a schematic diagram showing the temperature distribution of the electrochemical cell in a case where the first circulation operation and the second circulation operation are alternately repeated.

DETAILED DESCRIPTION OF THE INVENTION

First Embodiment

[0022] FIG. 1 is a diagram showing a water supply system **10** according to the first embodiment. The water supply system **10** includes an electrochemical stack **12**, a water circulation flow path **14**, a heat exchanger **16**, and a circulation pump **18**.
[0023] The electrochemical stack **12** is a stack capable of performing an electrochemical reaction. The electrochemical stack **12** may be a water electrolysis stack capable of electrolyzing water. Alternatively, the electrochemical stack **12** may be a hydrogen compression stack (an electrochemical hydrogen pump stack) capable of electrolyzing hydrogen (hydrogen gas). Alternatively, the electrochemical stack **12** may be a fuel cell stack capable of performing an electrochemical reaction between hydrogen (hydrogen gas) and oxygen (oxygen gas). The electrochemical stack **12** is formed in a substantially cylindrical shape, for example.
[0024] The electrochemical stack **12** is provided with a water introduction unit **20** for introducing water from the outside, and a water lead-out unit **22** for leading out the water introduced from the water introduction unit **20**. The water lead-out unit **22** includes a first water lead-out unit **24** and a second water lead-out unit **26**.
[0025] The water circulation flow path **14** is a channel for allowing water to flow through the electrochemical stack **12**. The water circulation flow path **14** is connected to the electrochemical stack **12**. The water circulation flow path **14** includes a first flow path portion **14a**, a second flow

path portion **14b**, and a third flow path portion **14c**. The first flow path portion **14a** connects the water introduction unit **20** and the circulation pump **18**. The second flow path portion **14b** connects the first water lead-out unit **24** and the circulation pump **18**. The third flow path portion **14c** connects the second water lead-out unit **26** and the second flow path portion **14b**. The third flow path portion **14c** may connect the second water lead-out unit **26** and the circulation pump **18**. In this case, the second flow path portion **14b** connects the first water lead-out unit **24** and the third flow path portion **14c**.

[0026] The heat exchanger **16** is provided at a portion between the circulation pump **18** and a connecting portion between the second flow path portion **14b** and the third flow path portion **14c** of the water circulation flow path **14**. The heat exchanger **16** cools water that is heated by heat generated in the electrochemical stack **12**. The heat exchanger **16** may be a radiator.

[0027] The circulation pump **18** circulates water between the heat exchanger **16** and the electrochemical stack **12** through the water circulation flow path **14**.

[0028] FIG. **2** is a schematic diagram showing the electrochemical stack **12** according to the first embodiment. The electrochemical stack **12** includes a plurality of electrochemical cells **30**, end plates **32a** and **32b**, a water passage member **34**, a water introducing communication passage **36**, and a water lead-out communication passage **38**.

[0029] The plurality of electrochemical cells **30** are stacked between the end plates **32a** and **32b**. The end plate **32a** is a plate that constitutes one end of the electrochemical stack **12** in the stacking direction of the electrochemical cells **30**. The end plate **32b** is a plate that constitutes the other end of the electrochemical stack **12** in the stacking direction of the electrochemical cells **30**.

[0030] The electrochemical cell **30** may be a water electrolysis cell capable of electrolyzing water. Alternatively, the electrochemical cell **30** may be a hydrogen compression cell (an electrochemical hydrogen pump cell) capable of electrolyzing hydrogen. Alternatively, the electrochemical cell **30** may be a unit cell capable of performing an electrochemical reaction between hydrogen and oxygen.

[0031] Although not shown in detail, a membrane electrode assembly is provided in each of the plurality of electrochemical cells **30**. The membrane electrode assembly includes an electrolyte membrane, and an anode current collector (an anode) and a cathode current collector (a cathode) disposed on both sides of the electrolyte membrane in the thickness direction thereof.

[0032] Each of the plurality of electrochemical cells **30** is provided with a first port **40**, a second port **42**, and a water supply path **44**. The first port **40** is connected to the water introducing communication passage **36**. The first port **40** is located on the opposite side to the water introduction unit **20** in a direction intersecting (an orthogonal direction to) the stacking direction of the electrochemical cells **30**. More specifically, the first port **40** is provided at a position shifted in phase by 180° in the circumferential direction of the electrochemical stack **12** from the water introduction unit **20**. The second port **42** is connected to the water lead-out communication passage **38**. The second port **42** is located on the opposite side to the first port **40** in the direction intersecting (the orthogonal direction to) the stacking direction of the electrochemical cells **30**. The water supply path **44** connects the first port **40** and the second port **42**. The water supply path **44** extends along the direction intersecting (the orthogonal direction to) the stacking direction of the electrochemical cells **30**. The number of the water supply paths **44** may be one or more.

[0033] The water passage member **34** is interposed between two electrochemical cells **30** located in a central region in the stacking direction, among the plurality of electrochemical cells **30**. No other electrochemical cell **30** is interposed between the two electrochemical cells **30** between which the water passage member **34** is interposed. The central region is, for example, a middle region obtained by dividing the plurality of electrochemical cells **30** into three equal parts in the stacking direction. The regions on the outer side in the stacking direction may be referred to as end regions. However, the central region may be, for example, a middle region obtained by dividing the plurality of electrochemical cells **30** into three parts at a ratio of 1:2:1 in the stacking direction.

[0034] In the present embodiment, the water passage member **34** is interposed between the electrochemical cell **30** located at one end in the stacking direction among the plurality of electrochemical cells **30**, and the electrochemical cell **30** located at the other end in the stacking direction among the plurality of electrochemical cells **30**, but the present invention is not limited thereto. Further, the water passage member **34** is formed in a plate shape (specifically, a disk shape), but the shape is not limited thereto.

[0035] Unlike the electrochemical cell **30**, the water passage member **34** does not include a membrane electrode assembly. Therefore, unlike the electrochemical cell **30**, electricity is not supplied to the water passage member **34**. The water passage member **34** is provided with the water introduction unit **20** and a flow path **50**. One end of the flow path **50** is connected to the water introduction unit **20**. The other end of the flow path **50** is connected to the water introducing communication passage **36**. The flow path **50** extends along the direction in which the water supply path **44** extends. The flow path **50** has a flow cross-sectional area larger than the flow cross-sectional area of the water supply path **44** of each of the plurality of electrochemical cells **30**, but the present invention is not limited thereto. The flow path **50** allows water introduced from the water introduction unit **20** to flow along the electrochemical cells **30** and guides the water to the water introducing communication passage **36**. The flow path **50** may be a groove formed in the water passage member **34**. In this case, the groove is sandwiched between two electrochemical cells **30**, thereby forming the flow path **50**.

[0036] The water introducing communication passage **36** extends along the stacking direction of the electrochemical cells **30**, and penetrates the plurality of electrochemical cells **30** and the water passage member **34** in the stacking direction. In other words, the water introducing communication passage **36** communicates with the flow path **50**. The water introducing communication passage **36** is disposed in the vicinity of the first port **40** of each of the plurality of electrochemical cells **30**, and guides water supplied from the flow path **50** to the first port **40** of each of the plurality of electrochemical cells **30**. That is, the water introducing communication passage **36** communicates with the first port **40** of each of the plurality of electrochemical cells **30** and the flow path **50**.

[0037] The water lead-out communication passage **38** extends along the stacking direction of the electrochemical cells **30**, and penetrates the plurality of electrochemical cells **30** in the stacking direction. The water lead-out communication passage **38** communicates with the second port **42** of each of the plurality of electrochemical cells **30**, but does not communicate with the flow path **50**. It should be noted that the water lead-out communication passage **38** may penetrate the water passage member **34** as long as the water lead-out communication passage **38** does not communicate with the flow path **50**. The water lead-out communication passage **38** is disposed in the vicinity of the second port **42** of each of the electrochemical cells **30**, and guides water discharged from the second port **42** of each of the electrochemical cells **30** to the water lead-out unit **22** (the first water lead-out unit **24** and the second water lead-out unit **26**). The first water lead-out unit **24** is provided in the electrochemical cell **30** located at one end in the stacking direction, among the plurality of electrochemical cells **30**. The second water lead-out unit **26** is provided in the electrochemical cell **30** located at the other end in the stacking direction, among the plurality of electrochemical cells **30**.

[0038] In the water supply system **10** configured as described above, the circulation pump **18** is driven during the operation of the electrochemical stack **12**. Water is circulated as follows in accordance with the driving of the circulation pump **18**.

[0039] As shown in FIG. **1**, the water output from the circulation pump **18** is supplied to the water introduction unit **20** (into the electrochemical cell **30** located at substantially the center in the stacking direction of the electrochemical cells **30**) through the first flow path portion **14a**. As shown in FIG. **2**, the water supplied to the water introduction unit **20** flows through the flow path **50** of the water passage member **34**. The water flowing through the flow path **50** passes between the two electrochemical cells **30** located in the central region in the stacking direction of the

electrochemical cells **30**, and flows into the water introducing communication passage **36**. The water that has flowed into the water introducing communication passage **36** flows in the stacking direction of the electrochemical cells **30**, and reaches the first port **40** of each of the plurality of electrochemical cells **30**. The water that has reached the first port **40** flows out from the second port **42** into the water lead-out communication passage **38** through the water supply path **44**. The water that has flowed out into the water lead-out communication passage **38** flows in the stacking direction of the electrochemical cells **30**, and reaches the first water lead-out unit **24** or the second water lead-out unit **26**. As shown in FIG. **1**, the water that has reached the first water lead-out unit **24** is supplied to the circulation pump **18** through the second flow path portion **14b**. On the other hand, the water that has reached the second water lead-out unit **26** flows into the second flow path portion **14b** through the third flow path portion **14c**, and is supplied to the circulation pump **18**.

[0040] In response to the operation of the electrochemical stack **12**, the temperature of the electrochemical stack **12** increases. It has been found that, in the electrochemical stack **12**, the temperature of the electrochemical cells **30** increases as the electrochemical cells **30** are located closer to the middle in the stacking direction. This is because the electrochemical cells **30** located in the middle are less likely to dissipate heat than the electrochemical cells **30** located at the ends. [0041] The electrochemical stack **12** of the present embodiment is provided with the water passage member **34**. The water passage member **34** is interposed between two electrochemical cells **30** located in the central region in the stacking direction of the electrochemical cells **30**, among the plurality of electrochemical cells **30**. The water passage member **34** is provided with the flow path **50** for allowing water supplied from the outside to flow along the electrochemical cells **30** and guiding the water to the water introducing communication passage **36**. Therefore, the water supplied from the outside can be distributed and supplied to each of the plurality of electrochemical cells **30** after the entire amount of the water is passed between the electrochemical cells **30** located in the central region, which have a relatively high temperature. As a result, the amount of water flowing through the flow path **50** is larger than the amount of water flowing through each of the electrochemical cells **30**, and it is possible to increase the degree of cooling of the electrochemical cells **30** located in the central region, which have a relatively high temperature.

[0042] Further, in the present embodiment, in a case where water is distributed to each of the plurality of electrochemical cells **30** through the first port **40** of each of the electrochemical cells **30** communicating with the water introducing communication passage **36**, the first port **40** is located on the opposite side to the water introduction unit **20** for introducing the water supplied from the outside (see FIG. **2**). Therefore, it is easy to increase the distance for water to pass between the electrochemical cells **30** having a relatively high temperature.

[0043] Further, in the present embodiment, the flow path **50** has a flow cross-sectional area larger than the flow cross-sectional area of the water supply path **44** provided in each of the electrochemical cells **30**. Therefore, compared to a case where the cross-sectional area of the flow path **50** is equal to or less than the cross-sectional area of the water supply path **44**, the degree of cooling of the electrochemical cells **30** in the central region is easily increased. In particular, when viewed from the stacking direction, the flow cross-sectional area of the flow path **50** is preferably larger than the flow cross-sectional area of the water supply path **44**.

[0044] Furthermore, in the present embodiment, the water passage member **34** is interposed between the electrochemical cell **30** located at one end in the stacking direction of the electrochemical cells **30** among the plurality of electrochemical cells **30**, and the electrochemical cell **30** located at the other end in the stacking direction of the electrochemical cells **30** among the plurality of electrochemical cells **30**, in other words, between the electrochemical cells **30** in the end regions. Therefore, compared to a case where the water passage member **34** is interposed between other two electrochemical cells **30**, the degree of cooling of the electrochemical cells **30** in the central region is easily increased.

Second Embodiment

[0045] In the second embodiment, the same components as those described in the first embodiment are denoted by the same reference numerals. In the second embodiment, the description overlapping with that of the first embodiment will be omitted. FIG. 3 is a diagram showing the water supply system **10** according to the second embodiment.

[0046] In the present embodiment, the water supply system **10** further includes a plurality of on-off valves **60A**, **60B**, **60C**, **60D**, **60E**, and **60F**, and a control unit **62**. The plurality of on-off valves **60A**, **60B**, **60C**, **60D**, **60E**, and **60F** are provided in the water circulation flow path **14**. The control unit **62** drives the circulation pump **18** during the operation of the electrochemical stack **12**. The control unit **62** controls the plurality of on-off valves **60A**, **60B**, **60C**, **60D**, **60E**, and **60F** during the operation of the electrochemical stack **12**.

[0047] In the present embodiment, the electrochemical stack **12** includes two water introduction units **20**, two first water lead-out units **24**, and two second water lead-out units **26**. One of the two water introduction units **20** is referred to as a water introduction unit **20x**, and the other of the two water introduction units **20** is referred to as a water introduction unit **20y**. The water introduction unit **20y** is provided at a position shifted in phase by 180° in the circumferential direction of the electrochemical stack **12** from the water introduction unit **20x**. One of the two first water lead-out units **24** is referred to as a first water lead-out unit **24x**, and the other of the two first water lead-out units **24** is referred to as a first water lead-out unit **24y**. The first water lead-out unit **24y** is provided at a position shifted in phase by 180° in the circumferential direction of the electrochemical stack **12** from the first water lead-out unit **24x**. One of the two second water lead-out units **26** is referred to as a second water lead-out unit **26x**, and the other of the two second water lead-out units **26** is referred to as a second water lead-out unit **26y**. The second water lead-out unit **26y** is provided at a position shifted in phase by 180° in the circumferential direction of the electrochemical stack **12** from the second water lead-out unit **26x**.

[0048] In the present embodiment, the water circulation flow path **14** further includes a fourth flow path portion **14d**, a fifth flow path portion **14e**, and a sixth flow path portion **14f**. The fourth flow path portion **14d** connects the first flow path portion **14a** and the water introduction unit **20y**. The fifth flow path portion **14e** connects the second flow path portion **14b** and the first water lead-out unit **24y**. The sixth flow path portion **14f** connects the fifth flow path portion **14e** and the second water lead-out unit **26y**. The sixth flow path portion **14f** may connect the second flow path portion **14b** and the second water lead-out unit **26y**. In this case, the fifth flow path portion **14e** connects the sixth flow path portion **14f** and the second water lead-out unit **26y**.

[0049] The on-off valve **60A** is provided in the first flow path portion **14a** and near the water introduction unit **20x**. The on-off valve **60B** is provided in the fourth flow path portion **14d** and near the water introduction unit **20y**. The on-off valve **60C** is provided in the second flow path portion **14b** and near the first water lead-out unit **24x**. The on-off valve **60D** is provided in the fifth flow path portion **14e** and near the first water lead-out unit **24y**. The on-off valve **60E** is provided in the third flow path portion **14c** and near the second water lead-out unit **26x**. The on-off valve **60F** is provided in the sixth flow path portion **14f** and near the second water lead-out unit **26y**.

[0050] FIG. 4 is a schematic diagram showing the electrochemical stack **12** according to the second embodiment. In the present embodiment, the electrochemical stack **12** includes two water introducing/lead-out communication passages **64** instead of the water introducing communication passage **36** and the water lead-out communication passage **38**. One of the two water introducing/lead-out communication passages **64** is referred to as a water introducing/lead-out communication passage **64x**, and the other of the two water introducing/lead-out communication passages **64** is referred to as a water introducing/lead-out communication passage **64y**. The water introducing/lead-out communication passage **64x** and the water introducing/lead-out communication passage **64y** function as the water introducing communication passage **36** or the water lead-out communication passage **38**. In a case where the water introducing/lead-out communication passage **64x** functions as the water introducing communication passage **36**, the

water introducing/lead-out communication passage **64y** functions as the water lead-out communication passage **38**. Conversely, in a case where the water introducing/lead-out communication passage **64y** functions as the water introducing communication passage **36**, the water introducing/lead-out communication passage **64x** functions as the water lead-out communication passage **38**.

[0051] In the present embodiment, the configuration of the water passage member **34** is different from that of the first embodiment. FIG. **5** is a schematic diagram showing the configuration of the water passage member **34**. FIG. **5** shows the water passage member **34** as viewed from the front in the stacking direction of the electrochemical cells **30**. The water passage member **34** shown in FIG. **5** is formed in a disk shape, but the shape is not limited thereto.

[0052] The water passage member **34** is provided with the water introduction unit **20x**, the water introduction unit **20y**, and two flow paths **50**. One of the two flow paths **50** is referred to as a flow path **50x**, and the other of the two flow paths **50** is referred to as a flow path **50y**. One end of the flow path **50x** is connected to the water introduction unit **20x**. The other end of the flow path **50x** is connected to the water introducing/lead-out communication passage **64x**. One end of the flow path **50y** is connected to the water introduction unit **20y**. The other end of the flow path **50y** is connected to the water introducing/lead-out communication passage **64y**. The flow path **50x** and the flow path **50y** are provided separately in the water passage member **34** without communicating with each other.

[0053] In the water supply system **10** configured as described above, the control unit **62** alternately repeats a first circulation operation and a second circulation operation during the operation of the electrochemical stack **12**.

[0054] FIG. **6** is a diagram showing the flow of water in the water circulation flow path **14** during the first circulation operation. FIG. **7** is a diagram showing the flow of water in the electrochemical stack **12** during the first circulation operation. As shown in FIG. **6**, in the first circulation operation, the control unit **62** opens the on-off valves **60A**, **60C**, and **60E** and closes the on-off valves **60B**, **60D**, and **60F**.

[0055] The water output from the circulation pump **18** is supplied to the water introduction unit **20x** through the first flow path portion **14a**. As shown in FIG. **5**, the entire amount of the water supplied to the water introduction unit **20x** flows through the flow path **50x** of the water passage member **34**. As shown in FIG. **7**, the water flowing through the flow path **50x** passes between the two electrochemical cells **30** located in the central region in the stacking direction of the electrochemical cells **30**, and flows into the water introducing/lead-out communication passage **64x**. The water introducing/lead-out communication passage **64x** functions as the water introducing communication passage **36**. Specifically, the water flowing into the water introducing/lead-out communication passage **64x** flows in the stacking direction of the electrochemical cells **30**, and reaches the first port **40** of each of the plurality of electrochemical cells **30**. The water that has reached the first port **40** flows out from the second port **42** into the water introducing/lead-out communication passage **64y** through the water supply path **44**. The water introducing/lead-out communication passage **64y** functions as the water lead-out communication passage **38**.

Specifically, the water that has flowed out into the water introducing/lead-out communication passage **64y** flows in the stacking direction of the electrochemical cells **30**, and reaches the first water lead-out unit **24x** or the second water lead-out unit **26x**. As shown in FIG. **6**, the water that has reached the first water lead-out unit **24x** is supplied to the circulation pump **18** through the second flow path portion **14b**. On the other hand, the water that has reached the second water lead-out unit **26x** flows into the second flow path portion **14b** through the third flow path portion **14c**, and is supplied to the circulation pump **18**.

[0056] FIG. **8** is a diagram showing the flow of water in the water circulation flow path **14** during the second circulation operation. FIG. **9** is a diagram showing the flow of water in the electrochemical stack **12** during the second circulation operation. As shown in FIG. **8**, in the second

circulation operation, the control unit **62** opens the on-off valves **60B**, **60D**, and **60F** and closes the on-off valves **60A**, **60C**, and **60E**.

[0057] The water output from the circulation pump **18** flows into the fourth flow path portion **14d** through the first flow path portion **14a**, and is supplied to the water introduction unit **20y**. As shown in FIG. 5, the entire amount of the water supplied to the water introduction unit **20y** flows through the flow path **50y** of the water passage member **34**. As shown in FIG. 9, the water flowing through the flow path **50y** passes between the two electrochemical cells **30** located in the central region in the stacking direction of the electrochemical cells **30**, and flows into the water introducing/lead-out communication passage **64y**. The water introducing/lead-out communication passage **64y** functions as the water introducing communication passage **36**. Specifically, the water flowing into the water introducing/lead-out communication passage **64y** flows in the stacking direction of the electrochemical cells **30**, and reaches the second port **42** of each of the plurality of electrochemical cells **30**. The water that has reached the second port **42** flows out from the first port **40** into the water introducing/lead-out communication passage **64x** through the water supply path **44**. The water introducing/lead-out communication passage **64x** functions as the water lead-out communication passage **38**. Specifically, the water that has flowed out into the water introducing/lead-out communication passage **64x** flows in the stacking direction of the electrochemical cells **30**, and reaches the first water lead-out unit **24y** or the second water lead-out unit **26y**. As shown in FIG. 8, the water that has reached the first water lead-out unit **24y** flows into the second flow path portion **14b** through the fifth flow path portion **14e**, and is supplied to the circulation pump **18**. On the other hand, the water that has reached the second water lead-out unit **26y** flows into the fifth flow path portion **14e** through the sixth flow path portion **14f**, then flows into the second flow path portion **14b**, and is supplied to the circulation pump **18**.

[0058] As described above, in the electrochemical stack **12** according to the present embodiment, as in the first embodiment, the water supplied from the outside can be supplied to each of the plurality of electrochemical cells **30** after the water is passed between the electrochemical cells **30** that have a relatively high temperature. As a result, it is possible to increase the degree of cooling of the electrochemical cells **30** in the central region which have a relatively high temperature.

[0059] Further, in the electrochemical stack **12** according to the present embodiment, the first circulation operation and the second circulation operation are alternately repeated during the operation of the electrochemical stack **12**. In the first circulation operation, the control unit **62** adjusts the opening and closing of the plurality of on-off valves **60A** to **60F** so that the water flows from the first port **40** toward the second port **42** in each of the electrochemical cells **30**. On the other hand, in the second circulation operation, the control unit **62** adjusts the opening and closing of the plurality of on-off valves **60A** to **60F** so that the water flows from the second port **42** toward the first port **40** in each of the electrochemical cells **30**. Therefore, compared to a case where only the first circulation operation or the second circulation operation is performed, the difference between the upstream temperature and the downstream temperature of the water supply path **44** can be suppressed. Specifically, in a case where only the first circulation operation is performed, then as shown in FIG. 10A, the temperature in the vicinity of the second port **42** continues to be higher than the temperature in the vicinity of the first port **40**. On the other hand, in a case where only the second circulation operation is performed, then as shown in FIG. 10B, the temperature in the vicinity of the first port **40** continues to be higher than the temperature in the vicinity of the second port **42**. That is, the electrolysis efficiency and the progress of deterioration vary in the plane of the membrane electrode assembly. By alternately repeating the first circulation operation and the second circulation operation during the operation of the electrochemical stack **12**, the difference between the upstream temperature and the downstream temperature can be suppressed as shown in FIG. 10C. As a result, it is possible to suppress variations in electrolysis efficiency and variations in the progress of deterioration in the plane of the membrane electrode assembly. It should be noted that FIG. 10A, FIG. 10B, and FIG. 10C show the temperature distribution of the electrochemical

cell **30** when the electrochemical cell **30** is viewed from the stacking direction. In addition, FIG. **10A**, FIG. **10B**, and FIG. **10C** show the case where the number of the water supply paths **44** provided in the electrochemical cell **30** is plural.

[0060] The following supplementary notes are further disclosed in relation to the above-described embodiments.

Supplementary Note 1

[0061] The electrochemical stack (**12**) of the present disclosure includes: the plurality of electrochemical cells (**30**) stacked on one another; the water introduction unit (**20**) configured to introduce water supplied from the outside; the water lead-out unit (**22**) configured to lead out the water to the outside; the water introducing communication passage (**36**) penetrating the plurality of electrochemical cells along the stacking direction of the electrochemical cells, and configured to guide the water to the first port (**40**) provided in each of the plurality of electrochemical cells; the water lead-out communication passage (**38**) penetrating the plurality of electrochemical cells along the stacking direction, and configured to guide, to the water lead-out unit, the water discharged from the second port (**42**) provided in each of the plurality of electrochemical cells; and the water passage member (**34**) interposed between two of the plurality of electrochemical cells, the two electrochemical cells being located in the central region in the stacking direction, wherein the water passage member is provided with the water introduction unit, and the flow path (**50**) through which the entire amount of the water introduced from the water introduction unit flows along the electrochemical cells and is guided to the water introducing communication passage.

Supplementary Note 2

[0062] In the electrochemical stack according to Supplementary Note 1, the first port may be located on an opposite side to the water introduction unit in a direction intersecting the stacking direction.

Supplementary Note 3

[0063] In the electrochemical stack according to Supplementary Note 1 or 2, the flow cross-sectional area of the flow path may be larger than the flow cross-sectional area of the water supply path (**44**) that connects the first port and the second port.

Supplementary Note 4

[0064] In the electrochemical stack according to any one of Supplementary Notes 1 to 3, the water passage member may be interposed between the electrochemical cell located at one end in the stacking direction among the plurality of electrochemical cells and the electrochemical cell located at the other end in the stacking direction among the plurality of electrochemical cells.

Supplementary Note 5

[0065] In the electrochemical stack according to any one of Supplementary Notes 1 to 4, the water lead-out unit may include the first water lead-out unit (**24**) provided in the electrochemical cell located at one end in the stacking direction among the plurality of electrochemical cells, and the second water lead-out unit (**26**) provided in the electrochemical cell located at the other end in the stacking direction among the plurality of electrochemical cells.

Supplementary Note 6

[0066] In the electrochemical stack according to any one of Supplementary Notes 1 to 5, each of the plurality of electrochemical cells may be a water electrolysis cell configured to electrolyze the water.

[0067] Although the present disclosure has been described in detail, the present disclosure is not limited to the above-described individual embodiments. Various additions, replacements, modifications, partial deletions, and the like can be made to these embodiments without departing from the gist of the present disclosure, or without departing from the essence of the present disclosure derived from the claims and equivalents thereof. Further, these embodiments can also be implemented in combination. For example, in the above-described embodiments, the order of operations and the order of processes are shown as examples, and are not limited to these.

Furthermore, the same applies to a case where numerical values or mathematical expressions are used in the description of the above-described embodiments.

Claims

1. An electrochemical stack comprising: a plurality of electrochemical cells stacked on one another; a water introduction unit configured to introduce water supplied from an outside; a water lead-out unit configured to lead out the water to the outside; a water introducing communication passage penetrating the plurality of electrochemical cells along a stacking direction of the electrochemical cells, and configured to guide the water to a first port provided in each of the plurality of electrochemical cells; a water lead-out communication passage penetrating the plurality of electrochemical cells along the stacking direction, and configured to guide, to the water lead-out unit, the water discharged from a second port provided in each of the plurality of electrochemical cells; and a water passage member interposed between two of the plurality of electrochemical cells, the two electrochemical cells being located in a central region in the stacking direction, wherein the water passage member is provided with the water introduction unit, and a flow path through which an entire amount of the water introduced from the water introduction unit flows along the electrochemical cells and is guided to the water introducing communication passage.
 2. The electrochemical stack according to claim 1, wherein the first port is located on an opposite side to the water introduction unit in a direction intersecting the stacking direction.
 3. The electrochemical stack according to claim 1, wherein a flow cross-sectional area of the flow path is larger than a flow cross-sectional area of a water supply path that connects the first port and the second port.
 4. The electrochemical stack according to claim 1, wherein the water passage member is interposed between the electrochemical cell located at one end in the stacking direction among the plurality of electrochemical cells, and the electrochemical cell located at another end in the stacking direction among the plurality of electrochemical cells.
 5. The electrochemical stack according to claim 1, wherein the water lead-out unit includes a first water lead-out unit provided in the electrochemical cell located at one end in the stacking direction among the plurality of electrochemical cells, and a second water lead-out unit provided in the electrochemical cell located at another end in the stacking direction among the plurality of electrochemical cells.
 6. The electrochemical stack according to claim 1, wherein each of the plurality of electrochemical cells is a water electrolysis cell configured to electrolyze the water.
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